EV Battery Course — Module 2: Electrochemistry and Battery Fundamentals

Oct 9, 2026 · @Ashishpreet

A battery is a controlled chemical reaction split in two, so that the electrons have to travel through your motor to complete it. This module builds that idea from atoms up to the Nernst equation.

1. Learning objectives

- Explain oxidation and reduction, and how a cell separates them.
- Name the anode, cathode, electrolyte and separator, and say what each does.
- Explain why electrons flow through the wire but ions flow through the cell.
- Explain where cell voltage comes from, and why it falls under load.
- Explain intercalation, and why it makes lithium-ion cells long-lived.
- Calculate theoretical capacity, coulombic efficiency and round-trip efficiency.
- Use Faraday's law and the Nernst equation at an introductory level.

2. Fundamentals from first principles

Atoms, ions, electrons. An atom has a positive nucleus and negative electrons around it. Lose an electron and it becomes a positive ion ($Li \rightarrow Li^+ + e^-$). Charge is measured in coulombs; one electron carries 1.602×10^{-19} C. One mole of electrons carries 96,485 C, called the Faraday constant (F).

Oxidation and reduction (redox). Oxidation is losing electrons; reduction is gaining them. Memory aid: OIL RIG (Oxidation Is Loss, Reduction Is Gain). They always happen together; electrons lost by one species are gained by another.

The battery trick. If you mix the two reactants directly, electrons jump across and the energy appears as heat. A battery puts the two half-reactions in separate electrodes. The electrolyte lets ions pass but blocks electrons, so the electrons must take the external route through the load. That external flow is the useful current.

The four parts of a cell

Part	Job	In a typical Li-ion EV cell
Negative electrode (anode on discharge)	Releases electrons and Li^+ during discharge	Graphite on copper foil
Positive electrode (cathode on discharge)	Accepts electrons and Li^+ during discharge	LFP or NMC on aluminium foil
Electrolyte	Carries ions, blocks electrons	$LiPF_6$ salt in carbonate solvents
Separator	Keeps electrodes apart, lets electrolyte through	Porous polyethylene/polypropylene film, about 10–25 µm

Convention: in Li-ion engineering, "anode" always means the graphite (negative) electrode and "cathode" the metal-oxide (positive) one, named for discharge. Strictly, the roles swap during charging, but industry keeps the discharge names.

3. Detailed technical explanation

Electrode potential. Each electrode material holds lithium with a different "eagerness," measured as a potential against lithium metal (V vs Li/Li^+). Graphite holds lithium at about 0.1 V vs Li. LFP holds it at about 3.4 V and layered oxides such as NMC at about 3.7–3.9 V. Cell voltage is the difference: LFP–graphite ≈ 3.3 V, NMC–graphite ≈ 3.6–3.7 V.

Intercalation. Lithium ions slot between layers or into channels of a host crystal without destroying it, like books going onto a shelf. Charging pushes Li^+ out of the cathode and into the graphite (intercalation into the anode); discharging reverses it (deintercalation from the anode). Because the host structures survive, the reaction is highly reversible. That is why Li-ion cells last thousands of cycles while lead-acid plates, which are dissolved and rebuilt each cycle, last hundreds.

Reversible vs irreversible reactions. The intercalation reactions are reversible. Side reactions are not. The most important is the solid electrolyte interphase (SEI): on the first charge the electrolyte decomposes on graphite and forms a thin passivating film. It consumes some lithium permanently (first-cycle loss, typically about 5–10%) but then protects the anode. Its slow growth over life is a major ageing mechanism (Module 10).

OCV, loaded voltage and polarisation. Open-circuit voltage (OCV) is the voltage at rest, set by thermodynamics and state of charge. Under current, the terminal voltage moves away from OCV because of polarisation, which has three parts:

Loss	Cause	Timescale
Ohmic	Resistance of electrolyte, foils, contacts	Instant
Activation (charge transfer)	Energy barrier to move Li^+ across the electrode surface	Milliseconds
Concentration (diffusion)	Li^+ cannot diffuse fast enough; gradients build	Seconds to minutes

On discharge V_terminal < OCV; on charge V_terminal > OCV. The gap is lost as heat and is why fast charging and hard acceleration warm the pack.

Capacity, energy and power density

- Capacity (Ah): charge stored.
- Specific energy (Wh/kg) and energy density (Wh/L): energy per mass and volume; these set range.
- Specific power (W/kg): rate per mass; sets acceleration and charge acceptance.
- There is always a trade-off: thick electrodes store more energy but have longer ion paths and less power.

4. Diagram: discharge and charge

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flowchart LR
  A["Graphite anode (−)<br/>LiC6 → C6 + Li⁺ + e⁻"] -- "e⁻ through motor" -->
C["LFP cathode (+)<br/>FePO4 + Li⁺ + e⁻ → LiFePO4"]
  A -. "Li⁺ through electrolyte and separator" .-> C
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During discharge, electrons and lithium ions both travel from anode to cathode, by different routes. During charging, the charger forces both back the other way.

5. Step-by-step working principle (LFP–graphite cell, discharging)

1. Lithium atoms sit between graphite layers (LiC_6).
2. At the graphite surface each lithium gives up an electron (oxidation) and becomes Li^+.
3. The electron enters the copper current collector and travels through the external circuit, doing work.
4. The Li^+ moves into the electrolyte, through the separator's pores, toward the cathode.
5. At the cathode surface Li^+ and the returning electron meet. Fe^{3+} in $FePO_4$ is reduced to Fe^{2+}, and lithium enters the olivine crystal to make $LiFePO_4$.

6. As the anode empties and the cathode fills, the voltage difference falls until the BMS cut-off (about 2.5 V for LFP).
7. Charging reverses every step; the charger supplies the energy.

Half-reactions (discharge):

$$\text{Anode: } \mathrm{LiC_6 \rightarrow C_6 + Li^+ + e^-} \qquad \text{Cathode: } \mathrm{FePO_4 + Li^+ + e^- \rightarrow LiFePO_4}$$

6. Engineering equations

Theoretical specific capacity of a material

$$Q_{th}\ (\text{mAh/g}) = \frac{n \times F}{3.6 \times M}$$

n = electrons transferred per formula unit, F = 96,485 C/mol, M = molar mass (g/mol). The 3.6 converts coulombs per gram to mAh/g.

Faraday's law (mass of material for a given charge)

$$m = \frac{Q \times M}{n \times F}$$

Q in coulombs (1 Ah = 3,600 C), m in grams.

Nernst equation

$$E = E^\circ - \frac{RT}{nF} \ln Q_r$$

E° = standard potential (V), R = 8.314 J/(mol·K), T in kelvin, Q_r = reaction quotient (ratio of product to reactant activities). At 25 °C, RT/F = 0.0257 V. It explains why OCV shifts with state of charge and temperature.

Terminal voltage

$$V_{terminal} = OCV - I \times R_{internal} \quad (\text{discharge})$$

Efficiencies

$$\eta_{coulombic} = \frac{Ah_{out}}{Ah_{in}} \qquad \eta_{round\text{-}trip} = \frac{Wh_{out}}{Wh_{in}}$$

Coulombic efficiency of a healthy Li-ion cell is above 99.9% per cycle. Round-trip energy efficiency is lower, typically about 90–97%, because charge goes in at a higher voltage than it comes out.

7. Worked examples

Example 1: theoretical capacity of LFP. $LiFePO_4$, M = 157.76 g/mol, n = 1.

- Q = 96,485 ÷ (3.6 × 157.76) = 96,485 ÷ 567.9 = **169.9 mAh/g**. Practical cells achieve about 150–160 mAh/g.

Example 2: theoretical capacity of graphite. Per LiC_6, the host C_6 has M = 72.06 g/mol, n = 1.

- Q = 96,485 ÷ (3.6 × 72.06) = 96,485 ÷ 259.4 = **372 mAh/g**.

Example 3: lithium needed per Ah (Faraday).

- 1 Ah = 3,600 C. Moles of e^- = 3,600 ÷ 96,485 = 0.0373 mol.
- Mass of Li = 0.0373 × 6.94 = **0.259 g of active lithium per Ah**. A 100 Ah cell cycles about 26 g of lithium (more is supplied to cover first-cycle loss).

Example 4: voltage under load. OCV = 3.30 V, R = 1.0 mΩ, I = 200 A.

- V = 3.30 − 200 × 0.001 = **3.10 V**. Heat = 200^2 × 0.001 = **40 W**.

Example 5: efficiencies. A cell takes 100.2 Ah and 340 Wh during charge; it returns 100.0 Ah and 320 Wh.

- Coulombic = 100.0 ÷ 100.2 = **99.8%**. Round-trip = 320 ÷ 340 = **94.1%**.

Example 6: Nernst temperature term. At 45 °C (318 K), RT/F = 8.314 × 318 ÷ 96,485 = **0.0274 V**, about 7% larger than at 25 °C. Temperature measurably shifts OCV, which is why BMS OCV tables are temperature-dependent.

8. Real-world EV examples

- **LFP flat voltage curve:** LFP sits near 3.3 V across most of its state of charge. Great for constant power delivery, but hard for the BMS to estimate SOC from voltage. That is why LFP cars (BYD, Tata's LFP models) periodically ask for a charge to 100% to recalibrate.
- **NMC sloping curve:** voltage drops steadily from about 4.2 V to 3.0 V, which makes voltage-based SOC estimation easier.
- **Cold weather:** ionic conductivity and charge-transfer rates fall sharply at low temperatures, so polarisation grows. EVs lose power and range in winter and limit fast charging until the pack warms.

9. Manufacturing applications

- **Formation** is where the SEI is deliberately grown under controlled slow charging. It is one of the most energy- and time-intensive steps in cell manufacturing (Module 12).
- **Moisture control:** water reacts with $LiPF_6$ to form hydrofluoric acid (HF), which attacks electrodes. This drives dry-room requirements, often dew points around −40 °C or drier.
- **Electrode loading:** coating weight (mg/cm²) sets capacity per area. Variation in coating weight shows up later as cell-to-cell capacity spread, which is why coating weight is a key SPC parameter.
- **N/P ratio:** the anode is designed with slightly more capacity than the cathode (N/P about 1.05–1.2) so lithium never has nowhere to go during charging, which would cause plating.

10. Common misconceptions

- **"Electrons travel through the electrolyte."** They must not; if they did, the cell would self-discharge. Only ions cross.
- **"The anode is always the negative terminal."** Strictly, anode means where oxidation happens, which swaps on charge. Li-ion convention fixes the names to the discharge roles.
- **"Lithium-ion cells contain lithium metal."** Normally they contain lithium only as ions in compounds. Lithium metal appears only as a fault (plating) or in next-gen lithium-metal cells.
- **"Coulombic efficiency of 99.9% means negligible loss."** Over 1,000 cycles, 0.1% per cycle compounds to substantial lithium loss. Small differences in CE predict large differences in life.
- **"OCV equals voltage under load."** Only at rest, and only after relaxation (often 30 minutes to hours).

11. Practical case study: high first-cycle loss on a new electrolyte batch

Problem. After switching electrolyte supplier, a cell line sees first-cycle coulombic efficiency drop from about 90% to about 86%, and graded capacity shifts down.

Investigation. Incoming Karl Fischer tests show the new electrolyte at a higher moisture level, still within the supplier's spec but above the old typical value. Higher water means more HF and a thicker, less stable SEI that consumes more lithium.

Corrective action. Tighten the incoming moisture limit, add a lot-release test, and require a certificate of analysis per drum. Re-validate formation protocol with the new supplier's material.

Lesson. Electrochemistry makes material chemistry a direct driver of yield and capacity. Incoming quality control is not a paperwork step.

12. Revision notes

- A cell separates oxidation and reduction; electrons go through the load, ions through the electrolyte.
- Cell voltage = cathode potential − anode potential.
- Intercalation keeps host structures intact, giving long life.
- SEI forms on first charge, consumes lithium, then protects the anode.
- Terminal voltage = OCV ± polarisation (ohmic, activation, concentration).
- Q_th = nF/(3.6M). 1 Ah needs about 0.26 g of cycling lithium.
- Coulombic efficiency > 99.9%; round-trip energy efficiency about 90–97%.

13. Glossary

Term	Meaning
Redox	Paired reduction and oxidation reactions
Intercalation	Insertion of ions into a host lattice without destroying it
SEI	Solid electrolyte interphase; passivating film on the anode
OCV	Open-circuit voltage, measured at rest
Polarisation	Departure of terminal voltage from OCV under current
Faraday constant (F)	96,485 C per mole of electrons
N/P ratio	Anode-to-cathode capacity ratio
Plating	Lithium depositing as metal on the anode surface
Coulombic efficiency	Charge out ÷ charge in

14. Knowledge check

Conceptual

1. Why must the electrolyte conduct ions but not electrons?
2. Why does an LFP–graphite cell have a lower voltage than an NMC–graphite cell?
3. Why is intercalation more reversible than the lead-acid reaction?
4. Name the three types of polarisation and one cause of each.
5. Why is the anode built with slightly more capacity than the cathode?

Numerical

1. Calculate the theoretical capacity of $LiCoO_2$ (M = 97.87 g/mol) assuming full delithiation (n = 1). Practical LCO uses only about half; what is that?
2. A 3.65 V, 60 Ah cell has 0.8 mΩ internal resistance and OCV 3.70 V. Find terminal voltage and heat at 180 A discharge.
3. A cell returns 99.95% coulombic efficiency per cycle with all loss being lithium loss. Roughly what capacity remains after 1,000 cycles?

Application

1. A supplier offers a cheaper electrolyte with 50 ppm more moisture. What tests would you require before approving it?
2. Customers of an LFP car complain that the range estimate jumps by 30 km after a full charge. Explain why, electrochemically.

15. Answer key

Conceptual

1. If electrons could cross inside, the reaction would proceed internally as heat (self-discharge or short circuit) instead of driving current through the load.
2. Cell voltage is the difference in electrode potentials. LFP holds lithium at about 3.4 V vs Li; NMC at about 3.7–3.8 V. Same anode, so NMC gives the higher difference.
3. In intercalation the host crystals stay intact; only lithium moves in and out. In lead-acid both electrodes are converted to lead sulphate and rebuilt each cycle, which coarsens crystals and sheds material.
4. Ohmic: electrolyte and contact resistance. Activation: energy barrier to charge transfer at the surface. Concentration: slow diffusion of Li^+ building up gradients.
5. So the graphite always has room for the incoming lithium during charge. Without spare capacity, lithium plates as metal on the anode, causing capacity loss and safety risk.

Numerical

1. Q = 96,485 ÷ (3.6 × 97.87) = 96,485 ÷ 352.3 = **273.9 mAh/g** theoretical. Practical (about half, to keep the structure stable) ≈ **137 mAh/g**, close to real LCO values of about 140 mAh/g.
2. V = 3.70 − 180 × 0.0008 = 3.70 − 0.144 = **3.556 V**. Heat = $180^2 \times 0.0008$ = **25.9 W**.
3. 0.9995^{1000} = e^(1000 × ln 0.9995) = e^(−0.50) ≈ **0.61, about 61% remaining**. This shows why CE must be extremely close to 100% (real cells also recover some and lose capacity by other routes, so this is a simplified model).

Application

1. Karl Fischer moisture on every lot; HF/acidity; build cells with it and compare first-cycle efficiency, capacity, impedance growth and gassing; accelerated cycle-life and high-temperature storage tests against the current electrolyte; a PPAP-style approval before any change. Weigh savings against yield loss.
2. LFP's voltage is nearly flat across most of the SOC range, so voltage tells the BMS little about SOC. The BMS relies on coulomb counting, which drifts. At full charge the voltage rises sharply, giving a reliable reference, so the BMS resets SOC and the estimate jumps.

16. Further learning

- *Linden's Handbook of Batteries* (McGraw-Hill): the standard reference.
- *Lithium-Ion Batteries: Basics and Applications*, Reiner Korthauer (Springer): materials to cells, industry-oriented.
- Battery University: articles on how Li-ion works and on SEI.
- MIT OpenCourseWare 10.626 Electrochemical Energy Systems (Martin Bazant): rigorous, free lectures.

Next: Module 3, lithium-ion chemistries in detail.